Kalpesh Khare

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Education

Indian Institute of Technology Bombay, India Expected May 2026
Bachelor of Technology ($GPA : 8.36/10.00$) *Mumbai, India*

- **Relevant Coursework:** Advanced Image Processing, Fundamentals of Digital Image Processing, Optimization in ML, Introduction to Machine Learning, AI and Data Science, Advanced Methods in Satellite Image Processing

Internship

Tonichi Insatsu - T-NEXT dept. (JAPAN) [Pre-Placement Offer] June 2025– July 2025
ML & Software Development Intern *Tokyo, Japan*

- Developed an end-to-end ML pipeline for business card digitization using OCR, image processing,& OpenAI APIs.
- Improved extraction and classification accuracy from 92.5% to 99.2% through model optimization and tuning.
- Solved challenges in Japanese script recognition, enhancing multilingual robustness of the system.
- Awarded a Pre-Placement Offer in recognition of impactful contributions and performance during the internship.

Tonichi Insatsu - T-NEXT dept. (remote) Dec 2025– March 2026
ML & Software Development Intern *Tokyo, Japan*

- Developing an end-to-end ML pipeline to summarize business negotiations from PDFs and audio attachments.
- Integrating the solution into a PHP Laravel application for seamless deployment and user interaction.

Projects

Adaptive Video Compressive Sensing using Reinforcement Learning
Guided by Prof. Ajit Rajwade, Dept. of CSE, IIT Bombay
| *Python, PyTorch, OpenCV, Reinforcement Learning, CNN, YOLO*

- Designed RL-based adaptive sensing framework to dynamically adjust video compression ratios for scene dynamics
- Developed deep learning models for video reconstruction with integrated object detection feedback
- Built an end-to-end pipeline combining compressive sensing, CNN reconstruction, and RL optimization

Image Restoration using Diffusion Models
Guided by Prof. Suyash P. Awate, Dept. of CSE, IIT Bombay
| *Python, PyTorch, OpenCV, Diffusion Models, CNN*

- Developed an image restoration pipeline using diffusion models for deblurring and super-resolution tasks
- Implemented iterative denoising with data-fidelity constraints for solving inverse imaging problems
- Designed sampling-based reconstruction methods improving robustness to noise and reconstruction quality

WeaSEL+RETRIEVE for Weak Supervision
Guided by Prof. Ganesh Ramakrishnan, Dept. of CSE, IIT Bombay
| *Python, PyTorch, Weak Supervision, Optimization*

- Engineered WeaSEL+RETRIEVE integration with Taylor-based coreset scoring, improving F1 by 9.5% & reducing error by 15%
- Built a noise-robust PyTorch pipeline with 99 weak labeling functions & gradient-based resolution, improving accuracy by 5.4%
- Automated scalable coreset selection from 1M unlabeled samples using a weak supervision framework

Technical Skills

Languages: Python, C++, JavaScript
ML / AI: PyTorch, TensorFlow, OpenCV, CNNs, Transformers, Diffusion Models
Computer Vision: Image Classification, Object Detection (YOLO), Image Restoration, OCR
Tools: Git, Docker, AWS, Flask, Node.js

Positions of Responsibility

Institute Sports Technical Head & Sports Web Convener - *Institute Sports Council, IIT Bombay* May'23 – Mar'25
Led 3 web conveners & developed SportsApp, for 20+ sports serving 13k+ students
Inter-IIT Sports Coordinator - *56th Inter IIT Sports Meet, IIT Bombay* Aug'23 – Dec'23
Managed tech, logistics and execution for 2000+ athletes and 6 major events
Mentor (ISMP & D-AMP) - *Institute Student Mentorship Program & Department Academic Mentorship Program* Jun'24 – Present
Mentored 12 + 6 students, guiding academics and easing transition to IIT Bombay